

DATA QUALITY REPORT

Dataset: Spotify Historical Charts (2017–2021)

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1. Introduction

This report documents the data quality assessment and cleaning process performed on the 'Spotify Historical Charts (2017–2021)' dataset as part of the GlobalGroove capstone project. The dataset captures Spotify's Top 200 and Viral 50 chart rankings across 70+ countries, published every 2–3 days from January 1, 2017 through December 30, 2021, sourced via Kaggle.

Due to the file size of approximately 3.5GB — estimated at 25–30 million raw rows — all cleaning was performed in Power BI using Power Query. This is an industry-standard approach for datasets of this scale and does not affect the validity of the cleaning steps performed.

This dataset will ultimately be merged with a second dataset (Top Spotify Songs in 73 Countries, 2023–2025) into a unified Microsoft SQL Server database. The cleaning steps documented here prepare Dataset 1 for that ingestion process.

2. Data Quality Issues Identified

The following issues were identified upon initial review of the raw CSV file in Power Query:

Issue Type	Column(s) Affected	Finding	Cleaning Action
Missing Values	region	NULL values present. This dataset does not include a Global Top 50 playlist in the same way as Dataset 2, but NULL region values indicate entries with no assigned country and should be preserved rather than discarded.	Filled NULL values with the string 'GLOBAL' to preserve these rows as a valid, queryable category. Consistent with the treatment of NULL country values in Dataset 2.
Missing Values	streams	NULL values present on all Viral 50 chart entries. This is expected and documented behavior — Spotify does not publish stream counts for the Viral 50 chart.	Retained NULLs. Documented as an expected condition, not a data error. Filling with zero would be factually incorrect, as zero streams is meaningfully different from no streams data.
Incorrect Data Type	date	Column stored as plain text string in the raw CSV despite containing date values.	Converted to Date type in Power Query. No errors encountered —

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			date formatting was consistent throughout the dataset.
Packed / Delimited Data	artist	Multiple artists stored in a single comma-space delimited string (e.g. 'Drake, 21 Savage'). This violates first normal form and prevents artist-level querying.	Split on delimiter ', ' using Power Query Split Column. Produced 4 artist columns, indicating some tracks have up to 4 credited artists. Columns renamed to artist_primary through artist_4. Full relational split into ArtistSong join table will be completed at SQL ingestion.
Duplicate Rows	title + date + region + chart	Potential duplicate entries where the same song appears more than once for the same region on the same date on the same chart type. The chart column is included in the composite key because the same song can legitimately appear on both the Top 200 and Viral 50 on the same date — those are two distinct chart events.	Removed duplicates using the composite key: title + date + region + chart. This combination uniquely identifies one chart event. Cross-region and cross-date appearances of the same track are valid and were preserved.

3. Cleaning Steps Performed

In Power BI — Power Query:

- Imported raw CSV via Get Data > Text/CSV. Selected 'Transform Data' to open Power Query before loading, preventing uncleaned data from entering the report.
- Replaced NULL values in the 'region' column with the string 'GLOBAL' using Replace Values.
- Converted 'date' from Text to Date type. No errors encountered.
- Removed duplicate rows on the composite key: title + date + region + chart. Four columns are required (rather than three as in Dataset 2) because Dataset 1 contains two chart types — Top 200 and Viral 50 — and the same song may legitimately appear on both on the same date.
- Split the 'artist' column by the delimiter ', ' (comma-space) at each occurrence, producing 4 positional artist columns.
- Renamed resulting columns to: artist_primary, artist_2, artist_3, artist_4. The first position is designated primary as it consistently represents the lead artist in the source data. The maximum number of credited artists observed in this dataset is 4.
- Applied 'Close & Apply' to load the fully cleaned dataset into the Power BI report canvas.

4. Columns Added During SQL Ingestion

The following columns do not exist in the raw source data and will be added programmatically during the BULK INSERT process into Microsoft SQL Server. They are documented here for full traceability between the cleaning phase and the database schema.

Column	Derived / Hardcoded Value	Reason
source	'Kaggle_Historical'	This dataset will be merged with Dataset 2 (Spotify 73 Countries) into a shared ChartEntry table in SQL Server. The source column provides row-level traceability back to the originating dataset, which is critical for debugging ingestion issues and isolating data by source during analysis.
chart_type_id	FK to 'Top 200' or 'Viral 50' in ChartType lookup table — derived per row from the chart column value	Unlike Dataset 2 where chart_type_id was a single hardcoded constant (Top 50 only), Dataset 1 contains two chart types. The existing chart column value ('top200' or 'viral50') is mapped to the corresponding chart_type_id FK at ingestion.
spotify_id	Parsed from the url column — Spotify track ID extracted from the URL path at ingestion	Dataset 1 does not have a native spotify_id column. Dataset 2 did. The track ID is embedded in the url field (e.g. 'https://open.spotify.com/track/TRACKID') and is parsed at SQL ingestion to populate spotify_id on the Song table — the cross-dataset song matching key.
country_id	FK to Country lookup table — derived by matching the region full name to the Country.full_name column	Dataset 1 stores countries as full names (e.g. 'United States'). Dataset 2 used 2-letter ISO codes. The Country lookup table bridges both. At ingestion, each region value is matched against Country.full_name to assign the country_id FK. 'GLOBAL' values map to NULL country_id, consistent with Dataset 2 global entry handling.

5. Remaining Data Limitations

- Artist normalization is partially complete. The 'artist' column has been split into 4 positional columns in Power BI, but the full relational split — extracting unique artist names, assigning integer IDs, and populating an ArtistSong join table — will be completed during SQL ingestion. Name-level deduplication (e.g. inconsistent casing, encoding differences) is also deferred to the SQL layer.
- Song-level deduplication across both datasets is deferred to SQL ingestion. The same track may appear in both Dataset 1 and Dataset 2. The spotify_id field — parsed from the url column at ingestion — will serve as the merge key to ensure each unique track resolves to one row in the Song table. URLs that cannot be parsed cleanly will result in NULL spotify_id values on the Song table.
- The 'streams' column is NULL for all Viral 50 entries by design. Spotify does not publish stream count data for the Viral 50 chart. This is a documented behavior of the source data, not a data quality error. Stored procedures must never compare stream counts (Dataset 1 Top 200) against the Dataset 2 popularity score in the same calculation — they measure different things via incompatible methodologies.

- This dataset stores countries as full names (e.g. 'United States') rather than ISO codes. Mapping to the Country lookup table occurs at SQL ingestion. Any full names not found in the lookup table will require manual review before ingestion is complete.
- Audio feature data (danceability, energy, valence, tempo, etc.) is not present in this dataset. These columns exist only in Dataset 2 and will be stored in a separate AudioFeatures table. Any stored procedures using AudioFeatures must LEFT JOIN and handle NULLs for all Dataset 1 songs.
- There is a temporal gap between Dataset 1 (ends December 2021) and Dataset 2 (starts October 2023). Trend analysis spanning this 22-month gap is not possible with the current data sources and must be clearly noted in all time-series visualizations, per the data disclaimer on the Welcome Screen.
- The 'trend' column uses categorical string values (MOVE_UP, MOVE_DOWN, SAME, NEW_ENTRY) specific to Dataset 1. Dataset 2 uses numeric integer movement fields instead. These are stored in ChartEntry as separate nullable columns and are never merged. Trend string normalization to numeric equivalents occurs at SQL ingestion.